

ON SOME NONCENTRAL DISTRIBUTIONS IN MULTIVARIATE ANALYSIS

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1. Introduction

Krishnaiah and Chang (1970, 1971) and Krishnaiah and Waikar (1971) derived exact expressions for the marginal distributions of any single or few ordered roots of a class of random matrices as well as the distributions of the traces of two random matrices. They derived the above results by exploiting the method of integration over alternate variables, Mehta (1967). For a very brief summary of the literature on the distributions of the individual roots and traces of some random matrices the reader is referred to the first three references above.

In general any suitable function(s) (not necessarily symmetric) of the roots of the appropriate random matrices can be used to test various hypotheses that arise in such problems as MANOVA, canonical correlation, tests for equality of covariance matrices, etc. Of course, the choice of these functions depends upon such factors as the optimum properties of the tests and the feasibility of the evaluation of the distributions. It will be of interest to examine the distributions of some of these functions which have at least intuitive appeal in testing some of the hypotheses.

In this note, we extend the results in Krishnaiah and Chang (1970, 1971) and Krishnaiah and Waikar (1971) to the non-central cases by using the method of de Bruijn (1955) for the evaluation of a certain integral. Taking advantage of de Bruijn's method, we have also derived the moments of the elementary symmetric functions of roots of a class of random matrices in the non-central case. Here we note that Pillai and his associates (see (1968a), (1968b) and (1969) and the references there) derived the moments of elementary symmetric functions of roots explicitly in some special cases.

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2. Preliminaries

De Bruijn (1955) proved the following useful result:*

Lemma 2.1 (de Bruijn)

Let (a, b) be any interval finite or infinite. Then

$$(2.1) \quad \int_{a \leq x_1 \leq \dots \leq x_n \leq b} \phi(x_1, \dots, x_n) dx_1 \dots dx_n = Pf(A)$$

where

$$\phi(x_1, \dots, x_n) = |y_{ij}|, \quad y_{ij} = \phi_i(x_j),$$

$A = (a_{ij})$ and $Pf(A)$ denotes the Pfaffian of A . Here the elements a_{ij} are given by

$$(2.2) \quad a_{ij} = \int_a^b \int_a^b \phi_i(x) \phi_j(y) \operatorname{Sgn}(y-x) dx dy \quad i, j = 1, \dots, 2m$$

if $n = 2m$; if $n = 2m+1$, then they are given by (2.2) and by

$$a_{2m+2, 2m+2} = 0,$$

$$a_{i, 2m+2} = -a_{2m+2, i} = \int_a^b \phi_i(x) dx \quad i = 1, \dots, 2m+1.$$

We need the following in the sequel:

Lemma 2.2

Let $\eta(x_1, \dots, x_n)$ be a symmetric function of $x_1, \dots, x_n$ and $\Phi(x_1, \dots, x_n)$ be as defined before. Then

$$(2.3) \quad \int_{a \leq x_1 \leq \dots \leq x_n \leq b} \eta(x_1, \dots, x_n) \phi(x_1, \dots, x_n) dx_1 \dots dx_n \\ = \int_a^b \dots \int_a^b E(x_1, \dots, x_n) \eta(x_1, \dots, x_n) \prod_{i=1}^n \{\phi_i(x_i) dx_i\}$$

where

$$E(x_1, \dots, x_n) = \prod_{1 \leq j < i \leq n} \operatorname{Sgn}(x_i - x_j)$$

In addition, if

$$(2.4) \quad \eta(x_1, \dots, x_n) = \sum_l C_l x_1^{k_l} \dots x_n^{k_n},$$

* The integral in Eq. (2.2) of Krishnaiah and Waikar (1971) is a special case of the integral in Eq. (2.1) of this paper.

where C_κ depends upon $k_1, \dots, k_n$ and Σ_1 denotes the summation over the different elements of $\kappa = (k_1, \dots, k_n)$ subject to suitable restrictions, we have

$$(2.5) \quad \int_a^b \dots \int_a^b \eta(x_1, \dots, x_n) \phi(x_1, \dots, x_n) dx_1 \dots dx_n \\ = \Sigma_1 C_\kappa Pf(A\kappa)$$

where $A\kappa = (a_{ij}^\kappa)$.

In the above lemma, a_{ij}^κ are given by

$$(2.6) \quad a_{ij}^\kappa = \int_a^b \int_a^b x^{k_i} \phi_i(x) y^{k_j} \phi_j(y) \text{Sgn}(y-x) dx dy \quad i, j=1, \dots, 2m$$

when $n = 2m$; if $n = 2m+1$, the elements a_{ij}^κ are given by (2.6) and

$$a_{2m+2, 2m+2}^\kappa = 0$$

$$a_{i, 2m+2}^\kappa = -a_{i, 2m+2}^\kappa = \int_a^b x^{k_i} \phi_i(x) dx \quad i=1, \dots, 2m+1.$$

Lemma 2.2 can be proved by following the same lines as in de Bruijn (1955).

3. Moments of Elementary Symmetric Functions of Roots

Let $\ell_1, \dots, \ell_p$ be the latent roots of a random matrix and let us consider the situations when their joint density in the non-central case is of the form

$$(3.1) \quad f(\ell_1, \dots, \ell_p) = C \prod_{i=1}^p \psi(\ell_i) \prod_{i>j} (\ell_i - \ell_j) \Sigma_2 \Sigma_1 a(\kappa) \eta_\kappa(L), \quad a \leq \ell_1 \leq \dots \leq \ell_p \leq b$$

where $L = \text{diag}(\ell_1, \dots, \ell_p)$, $\kappa = (k_1, \dots, k_p)$ is a partition of k subject to suitable restrictions, $\eta_\kappa(L)$ is a symmetric function and $a(\kappa)$ depends upon the population parameters and the elements of the partition κ . Also Σ_1 denotes the summation over all partitions of k , whereas Σ_2 denotes the summation over k . Now, let

$$(3.2) \quad \eta_\kappa(L) = \Sigma_3 b_r^\kappa \ell_1^{r_1} \dots \ell_p^{r_p}$$

where Σ_3 denotes the summation over $r_1, \dots, r_p$ subject to some suitable restrictions and b_r^κ depends on $r_1, \dots, r_p$ and κ . If $\eta_\kappa(L)$ is the zonal polynomial and the elements of the partition $\kappa = (k_1, \dots, k_p)$ are subject to the restriction $k_1 \geq \dots \geq k_p \geq 0$, then $\eta_\kappa(L)$ is of the form (3.2).

The elementary symmetric function of order q is given by

$$\zeta(\ell_1, \dots, \ell_p) = {}_{i_1 < \dots < i_q} \ell_{i_1} \dots \ell_{i_q} = \Sigma_4 \ell_{i_1} \dots \ell_{i_q} \text{ say.}$$

Now

$$(3.3) \quad \{\zeta(\ell_1, \dots, \ell_p)\}^s = \Sigma_5 (s_1, \dots, s_{p*}) \ell_1^{\eta_1} \dots \ell_p^{\eta_p},$$

where

$$p* = \binom{p}{q} \text{ and } \Sigma_5$$

denotes the summation over $s_1, \dots, s_{p*}$ such that $s_1 + \dots + s_{p*} = s$, $\eta_1 + \dots + \eta_p = qs$, and each η_i depends upon $s_1, \dots, s_{p*}$. We know that

$$\prod_{i=1}^p \psi(\ell_i) \eta_k(L) \{\zeta(\ell_1, \dots, \ell_p)\}^s$$

is a symmetric function of $\ell_1, \dots, \ell_p$. So by (2.3) and (2.5) we get the s -th moment of $\zeta(\ell_1, \dots, \ell_p)$ as

$$(3.4) \quad E[\{\zeta(\ell_1, \dots, \ell_p)\}^s] = C \Sigma_2 \Sigma_1 a(\kappa) \int_a^b \dots \int_a^b E(\ell_1, \dots, \ell_p) \\ \cdot \prod_{i=1}^p \{\psi(\ell_i) \ell_i^{i-1}\} \eta_k(L) \{\zeta(\ell_1, \dots, \ell_p)\}^s d\ell_1, \dots, d\ell_p \\ = C \Sigma_2 \Sigma_1 \Sigma_3 \Sigma_5 a(\kappa) b_r^k(s_1, \dots, s_{p*}) (A(r_1, \dots, r_p; \eta_1, \dots, \eta_p))$$

where

$$A(r_1, \dots, r_p; \eta_1, \dots, \eta_p) = \int_a^b \dots \int_a^b e(x_1, \dots, x_p) \prod_{i=1}^p \{\psi(x_i) x^{i+r_i + \eta_i - 1} dx_i\} \\ = \text{Pf}(a_{ij}(r_1, \dots, r_p; \eta_1, \dots, \eta_p))$$

and

$$a_{ij}(r_1, \dots, r_p; \eta_1, \dots, \eta_p) = \int_a^b \dots \int_a^b x^{i+r_i + \eta_i - 1} y^{j+r_j + \eta_j - 1} \\ \cdot \psi(x) \psi(y) \text{sgn}(y-x) dx dy$$

when $p=2m$; when $p=2m+1$, the elements $a_{ij}(r_1, \dots, r_p; \eta_1, \dots, \eta_p)$ are as given above and

$$a_{2m+2, 2m+2}(r_1, \dots, r_p; \eta_1, \dots, \eta_p) = 0 \\ a_{i, 2m+2}(r_1, \dots, r_p; \eta_1, \dots, \eta_p) \\ = -a_{2m+2, i}(r_1, \dots, r_p; \eta_1, \dots, \eta_p) = \int_a^b x^{r_i + \eta_i + i - 1} \psi(x) dx, \quad i=1, \dots, 2m+1.$$

The summations $\Sigma_1, \Sigma_2, \Sigma_3, \Sigma_5$, are defined in (3.1), (3.2) and (3.3).

In the central case, let us assume that the joint density of $\ell_1, \dots, \ell_p$ is of the form

$$f(\ell_1, \dots, \ell_p) = C \prod_{i=1}^p \psi(\ell_i) \prod_{i>j} (\ell_i - \ell_j).$$

In this case

$$E \{ \zeta(\ell_1, \dots, \ell_p) \}^s = C \Sigma_5 (s_1, \dots, s_{p*}) \text{Pf}(A(0, \dots, 0; \eta_1, \dots, \eta_p))$$

where $A(0, \dots, 0; \eta_1, \dots, \eta_p)$ is given by (3.4)

4. Distribution of a few ordered roots

We need the following in the sequel.

Remark 4.1

Let (a, b) be any interval finite or infinite. Also, let

$$I(k_1, \dots, k_n; a, b) = \int_R \dots \int_R \Phi(x_1, \dots, x_n; k_1, \dots, k_n) dx_1 \dots dx_n$$

where

$$R: a \leq x_1 \leq \dots \leq x_n \leq b$$

and

$$\Phi(x_1, \dots, x_n; k_1, \dots, k_n) = \begin{vmatrix} \phi_{k_1}(x_1) & \dots & \phi_{k_1}(x_n) \\ \vdots & & \vdots \\ \phi_{k_n}(x_1) & \dots & \phi_{k_n}(x_n) \end{vmatrix}$$

Then

$$(4.1) \quad I(k_1, \dots, k_n; a, b) = |D(k_1, \dots, k_n)|^{\frac{1}{2}}$$

where

$$D(k_1, \dots, k_n) = (d_{ij}(k_1, \dots, k_n)),$$

$$(4.2) \quad d_{ij}(k_1, \dots, k_n) = \int_a^b \int_a^b \phi_{k_i}(x) \phi_{k_j}(y) \text{Sgn}(y-x) dx dy \quad i, j=1, \dots, 2m$$

when $n = 2m$; if $n = 2m+1$, we have, in addition to (4.2),

$$d_{2m+2, 2m+2}(k_1, \dots, k_n) = 0$$

$$d_{i, 2m+2}(k_1, \dots, k_n) = -d_{2m+2, i}(k_1, \dots, k_n) = \int_a^b \phi_{k_i}(x) dx, \quad i=1, \dots, 2m+1.$$

Formula (4.1) is implicit in lemma 2.1.

Let (a, b) be any interval finite or infinite and let

$$R^*: a \leq x_1 \leq \dots \leq x_r \leq x_{r+1} \leq x_{r+s+1} \leq \dots \leq x_p \leq b$$

Then

$$\begin{aligned}
 (4.3) \int \dots \int_{R^*} \Phi(x_1, \dots, x_p) \eta(x_1, \dots, x_p) dx_1 \dots dx_r dx_{r+s+1} \dots dx_p \\
 = \Sigma_6 \Sigma_7 (-1)^{\frac{r(r+1)}{2} + \frac{s(s+1)}{2} + \sum_{i=1}^r k_i + \sum_{i=1}^s \alpha_i} \\
 \cdot V(x_{r+1}, \dots, x_{r+s}; \alpha_1, \dots, \alpha_s) \int \dots \int_{R^{**}} E(x_1, \dots, x_r) E(x_{r+s+1}, \dots, x_p) \\
 \cdot V(x_1, \dots, x_r; k_1, \dots, k_r) V(x_{r+s+1}, \dots, x_p; \beta_{r+s+1}, \dots, \beta_p) \\
 \cdot \eta(x_1, \dots, x_p) dx_1 \dots dx_r dx_{r+s+1} \dots dx_p; a \leq x_{r+1} \leq \dots \leq x_{r+s} \leq b,
 \end{aligned}$$

and $(k_1 < \dots < k_r)$ is a subset of $(1, \dots, p)$, $(t_1 < \dots < t_{p-r})$ is its complementary set. Also $(\alpha_1 < \dots < \alpha_s)$ is a subset of $(t_1 < \dots < t_{p-r})$ and $(\beta_{r+s+1} < \dots < \beta_p)$ is the complementary subset of $(\alpha_1 < \dots < \alpha_s)$ with respect to $(t_1 < \dots < t_{p-r})$, Σ_6 denotes this summation over all possible $\binom{p}{r}$ choices of $(k_1, \dots, k_r)$ and Σ_7 stand for the summation over $\binom{p-r}{s}$ possible choices of $(\alpha_1 < \dots < \alpha_s)$,

$$V(x_1, \dots, x_r; k_1, \dots, k_r) = \det(\phi_{k_i}(x_j))_{i,j=1,\dots,r}$$

and

$$R^{**}: \{a \leq x_i \leq x_{r+1}, (i=1, \dots, r); x_{r+s} \leq x_j \leq b, (j=r+s+1, \dots, p)\}.$$

Formula (4.3) follows by repeated use of Laplace expansion of $\Phi(x_1, \dots, x_p)$ starting with the first r columns and then with the first s columns of the remaining determinant. Now again if $\eta(x_1, \dots, x_p)$ is of the form $\eta(x_1, \dots, x_p) = \Sigma_8 C_\delta x_1^{d_1} \dots x_p^{d_p}$ such that $\eta(x_1, \dots, x_p)$ is symmetric in $x_1, \dots, x_p$ and Σ_8 is similarly defined as in (2.4)

$$\begin{aligned}
 (4.4) \int \dots \int_{R^*} \phi(x_1, \dots, x_p) \eta(x_1, \dots, x_p) dx_1 \dots dx_r dx_{r+s+1} \dots dx_p \\
 = \Sigma_8 \Sigma_6 \Sigma_7 (-1)^{\frac{r(r+1)}{2} + \frac{s(s+1)}{2} + \sum_{i=1}^r k_i + \sum_{i=1}^s \alpha_i} C_\delta
 \end{aligned}$$

$$\cdot V(x_{r+1}, \dots, x_{r+s}; \alpha_1, \dots, \alpha_s) x_{r+1}^{d_{r+1}} \dots x_{r+s}^{d_{r+s}} \\ \cdot I(k_1, \dots, k_r; d_1, \dots, d_r; a, x_{r+1}) I(\beta_{r+s+1}, \dots, \beta_p; d_{r+s+1}, \dots, d_p; x_{r+s}, b)$$

where Σ_8 is given in (2.4) and $I(k_1, \dots, k_r; d_1, \dots, d_r; a, b)$ is obtained from $I(k_1, \dots, k_r; a, b)$ replacing $\phi_{k_i}(x_j)$ with $x_j^{d_j} \phi_{k_i}(x_j)$.

Now let us take the joint density of the roots in the noncentral case as given by (3.1). Then applying formula (4.3) we get the density of roots $x_{r+1}, \dots, x_{r+s}$ as

$$f(x_{r+1}, \dots, x_{r+s}) \\ = C \Sigma_2 \Sigma_1 \Sigma_3 \Sigma_6 \Sigma_7 (-1)^{\frac{r(r+1)}{2} + \frac{s(s+1)}{2} + \sum_{i=1}^r k_i + \sum_{i=1}^s \alpha_i} \\ \cdot a(\kappa) b_{\lambda}^K I(k_1, \dots, k_r; \lambda_1, \dots, \lambda_r; a, x_{r+1}) I(\beta_{r+s+1}, \dots, \beta_p; \lambda_{r+s+1}, \dots, \lambda_p; x_{r+s}, b) \\ \cdot V(x_{r+1}, \dots, x_{r+s}; \alpha_1, \dots, \alpha_s) x_{r+1}^{d_{r+1}} \dots x_{r+s}^{d_{r+s}}, \quad a \leq x_{r+1} \leq \dots \leq x_{r+s} \leq b$$

where

$$(4.5) \quad \phi(x_1, \dots, x_p) = \det(x_j^{i-1})_{i,j=1,\dots,p}$$

$C, \Sigma_1, \Sigma_2, \Sigma_3, \Sigma_6, \Sigma_7$ are all defined earlier b_{λ}^K and $\lambda_1, \dots, \lambda_p$ are defined in the same way as $b_{r'}^K, r_1, \dots, r_p$ in (3.2).

5. Probability integrals of ordered roots

As before, let us assume that the joint density of the roots $\ell_1 < \dots < \ell_p$ is given by (3.1). Then the density functions of the extreme roots can be obtained by following the same lines as in Krishnaiah and Chang (1971) and using (2.3) and (2.5). Also the c.d.f. of the individual extreme root can be computed by applying (2.3) and (2.5). The probability integral associated with the joint density of any two ordered roots is easily obtained by following the same lines as in Krishnaiah and Waikar (1971) and applying formulae (2.3) and (2.5). The expressions associated with the density functions and c.d.f.s of the extreme roots and the probability integral associated with any two ordered roots are not given here for the sake of brevity. We will now derive the c.d.f.s associated with the intermediate roots.

It is known that

$$(5.1) \quad P[\ell_r \leq x] = P[\ell_{r+1} \leq x] + P[\ell_1 < \dots < \ell_r < x < \ell_{r+1} < \ell_p]$$

Now c.d.f.s of the extreme roots can be evaluated using (2.3) and (2.5). The second term in (5.1) can be calculated following similar lines as in Krishnaiah and Waikar (1971) and using (4.3) and (4.4). It is given by

$$\begin{aligned} & P[\ell_1 < \dots < \ell_r < x < \ell_{r+1} < \dots < \ell_p] \\ &= C \Sigma_2 \Sigma_1 \Sigma_3 \Sigma_6 (-1)^{\frac{1}{2}r(r+1) + \sum_{i=1}^r k_i} b_{\lambda}^k a(\kappa) \\ & \cdot I(k_1, \dots, k_r; \lambda_1, \dots, \lambda_r; a, x) I(t_1, \dots, t_{p-r}, t_{r+1}, \dots, \lambda_p; x, b) \end{aligned}$$

where

$$C, \Sigma_1, \Sigma_2, \Sigma_3, \Sigma_6, (k_1 < \dots < k_r), (t_1 < \dots < t_{p-r})$$

and

$$\Phi(\ell_1, \dots, \ell_r)$$

are as defined in (4.5).

6. Non Null Distributions of Traces

Here we use the results in Krishnaiah and Chang (1970) and treat both traces discussed there in a unified way. Let the joint density of the roots be given as (3.1). Then

$$E(e^{-t(\ell_1 + \dots + \ell_p)}) = C \Sigma_2 \Sigma_1 \Sigma_3 a(\kappa) b_r^k \text{Pf}(B_r)$$

where

$$B_r = (b_{r,ij})_{i,j=1,\dots,p}$$

and

$$b_{r,ij} = \int_a^b \int_a^b x^{r_1+i-1} y^{r_2+j-1} \psi(x) \psi(y) e^{-tx} e^{-ty} \text{sgn}(y-x) dx dy \quad i,j=1,\dots,p.$$

If p is odd, the B_r matrix has to be augmented as given in Section 2. Now to find the exact distribution we have to invert this Laplace transform. To this end we note that

$$\text{Pf}(B_r) = \Sigma^{\pm} b_{r,i_1,i_2} \dots b_{r,i_{2a-1},i_{2a}}$$

where the summation is over all possible choices of $i_1, \dots, i_{2a}$ subject to this restriction $i_1 < i_2, \dots, i_{2a-1} < i_{2a}$ and the sign is +ve or -ve according as the permutation is even or odd. Thus inverting the Laplace transform we get

$$f(V) = C \Sigma_2 \Sigma_1 \Sigma_3 \Sigma_6 a(\kappa) b_T h_T(i_1, \dots, i_{2a}; V)$$

where $h_T(i_1, \dots, i_{2a})$ is the inverse Laplace transform of $\text{Pf}(B_T)$. For discussion on inverse transform of the above type for two important cases we refer to Krishnaiah and Chang (1970).

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